RN SHETTY TRUST®

RNS INSTITUTE OF TECHNOLOGY

Affiliated to VTU, Recognized by GOK, Approved by AICTE, New Delhi
(NAAC 'A+ Grade' Accredited, NBA Accredited (UG - CSE, ECE, ISE, EIE and EEE)
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DEPARTMENT OF MCA

Web Technologies Laboratory (MMCL206)

Compiled by

Roopa H M
Assistant Professor
DEPARTMENT OF MCA
R N S Institute of Technology Bengaluru-98

Name: ______________________________

USN: ______________________________

Vision of the Department

Synergizing Computer Applications for Real World.

Mission of the Department

- Produce technologists of the highest caliber to engage in design research and development, to enable the nation to be self-reliant.

- Give conceptual orientation in basic computer applications and mathematics, motivate the students for lifelong learning.

- Integrate project experiences at every level of the postgraduate curriculum to give a firm practical foundation.

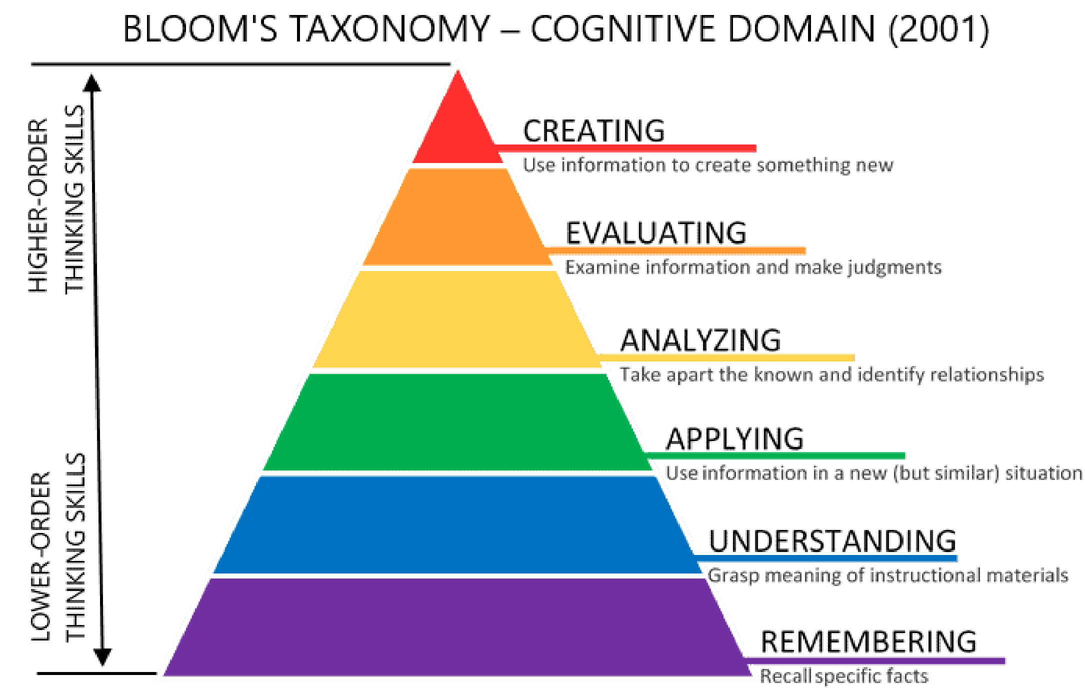
COURSE OUTCOMES

Course Outcomes: At the end of this course, students are able to:
CO1- To Provide a strong foundation in Web concepts, technology, and practice.
CO2- To Practice Web programming through a variety of programs.
CO3- To Understand the application of CSS..
CO4- To Demonstrate the use of concurrency and transactions in database.
CO5- Prepare students for industry roles involving data-driven decision making and predictive modeling.

COs and POs Mapping of lab Component

COURSE OUTCOMES	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	2	2	1	2	-	-	-	-	-	-	3	3	2	-	-
CO2	3	3	3	1	2	-	-	-	-	-	-	3	3	3	-	-
CO3	3	3	3	1	2	-	-	-	-	-	-	3	3	3	-	-
CO4	3	3	3	3	2	-	-	-	-	-	-	3	3	3	-	-

REVISED BLOOMS TAXONOMY (RBT)



PROGRAM LIST

Sl. NO.	*Program Description*	*Page No.*
1	Create an XHTML page that provides information about your department. Your XHTML page must use the following tags: a) Text Formatting tags b) Horizontal rule c) Meta element d) Links e) Images f) Tables (Use of additional tags encouraged).	
2	Develop and demonstrate a XHTML file that includes Javascript script for the following problems: a) Input : A number n obtained using prompt Output : The first n Fibonacci numbers b) Input : A number n obtained using prompt Output : A table of numbers from 1 to n and their squares using alert	
3	Develop and demonstrate, using JavaScript script, a XHTML document that contains three short paragraphs of text, stacked on top of each other, with only enough of each showing so that the mouse cursor can be placed over some part of them. When the cursor is placed over the exposed part of any paragraph, it should rise to the top to become completely visible. Modify the above document so that when a text is moved from the top stacking position, it returns to its original position rather than to the bottom	
4	Develop, test and validate an XHTML document that has checkboxes for apple (59 cents each), orange (49 cents each), and banana (39 cents each) along with submit button. Each check box should have its own onclick event handler. These handlers must add the cost of their fruit to the total cost. An event handler for the submit button must produce an alert window with the message „your total cost is $xxx", where xxx is the total cost of the chose fruit, including 5 percent sales tax. This handler must return „false" (to avoid actual submission of the form data). Modify the document to accept quantity for each item using textboxes.	
Extra practice Programs		
6		
7		
8		

Program 1

Create an XHTML page that provides information about your department. Your HTML page must use the following tags:

a) Text Formatting tags
b) Horizontal rule
c) Meta element
d) Links
e) Images
f) Tables (Use of additional tags encouraged).

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Department of MCA</title>
  </head>

  <body>
    <h1>Masters of Computer Application</h1>
            <hr/>
            <h2>About Us</h2>
    <pre>
              MCA is a post-graduate course.
              It is a two-year program spread over four semesters.
              The program is approved by AICTE and affiliated to VTU

    </pre>

     <h2>Research Areas</h2>
        <ul>
          <li>Artificial Intelligence</li>
          <li>Data Science</li>
          <li>Computer Vision</li>
          <li>Cyber Security</li>
        </ul>
```

```
<h2>Faculty Members</h2>

    <table border=1>
      <tr>
        <th>Name</th>
        <th>Designation</th>
        <th>Email</th>
      </tr>

      <tr>
        <td>    A    </td>
        <td>  Assistant Professor </td>
        <td>   a@gmail.com        </td>
      </tr>

      <tr>
        <td>    B   </td>
        <td>  Assistant Professor </td>
        <td>   b@gmail.com        </td>
      </tr>
    </table>


    <img src="D:\Web Technologies\Lab manual\college_logo.jpeg"/>

    <h2>Visit our College Website</h2>

    <a href="https://www.rnsit.ac.in/">College Website</a>

</body>
</html>
```

Output:

Masters of Computer Application

About Us

```
MCA is a post-graduate course.
It is a two-year program spread over four semesters.
The program is approved by AICTE and affiliated to VTU
```

Research Areas

- Artificial Intelligence
- Data Science
- Computer Vision
- Cyber Security

Faculty Members

Name	Designation	Email
A	Assistant Professor	a@gmail.com
B	Assistant Professor	b@gmail.com

Visit our College Website

College Website

Program 2

Develop and demonstrate a XHTML file that includes Javascript script for the following problems:

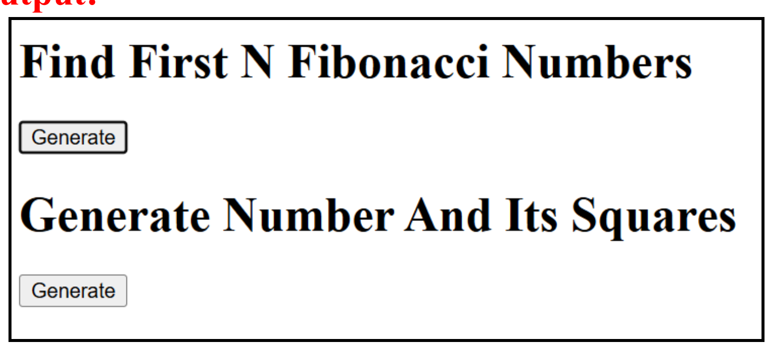
a) Input : A number n obtained using prompt Output : The first n Fibonacci numbers

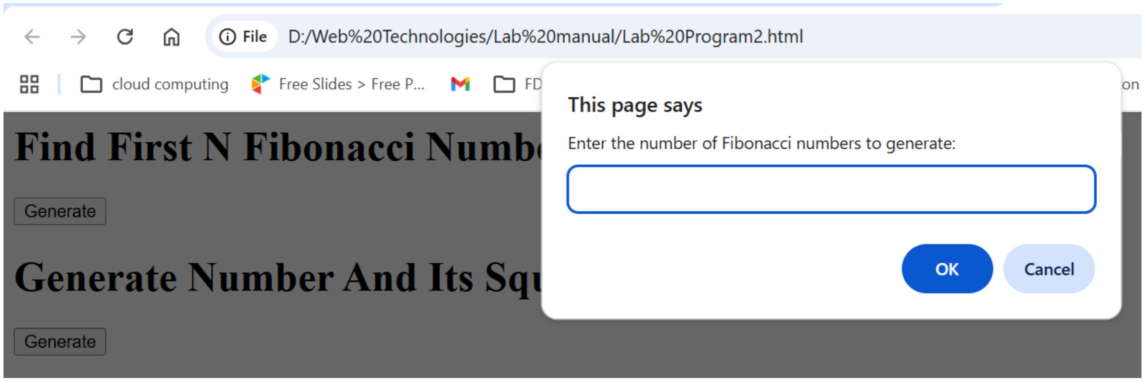
b) Input : A number n obtained using prompt Output : A table of numbers from 1 to n and their squares using alert

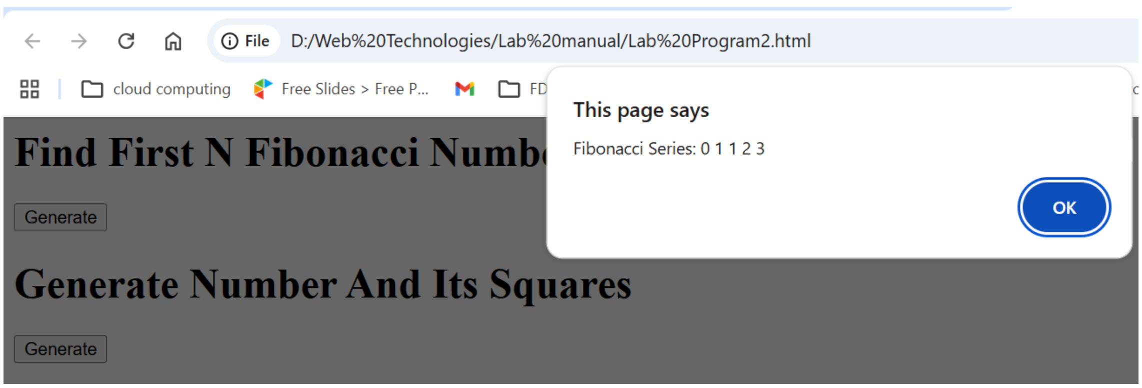
```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head>
  <title>JavaScript Problems - Fibonacci & Squares</title>

  <script type="text/javascript">
    // Function to generate Fibonacci series
    function fibonacci() {
      var n = prompt("Enter a number for Fibonacci series:", "5");
      n = parseInt(n);

      var a = 0, b = 1, c, result = "";

      if (n <= 0) {
        alert("Please enter a positive number");
        return;
      }

      result = "Fibonacci Series:\n";

      for (var i = 1; i <= n; i++) {
        result += a + " ";
        c = a + b;
        a = b;
        b = c;
      }

      alert(result);
    }
```

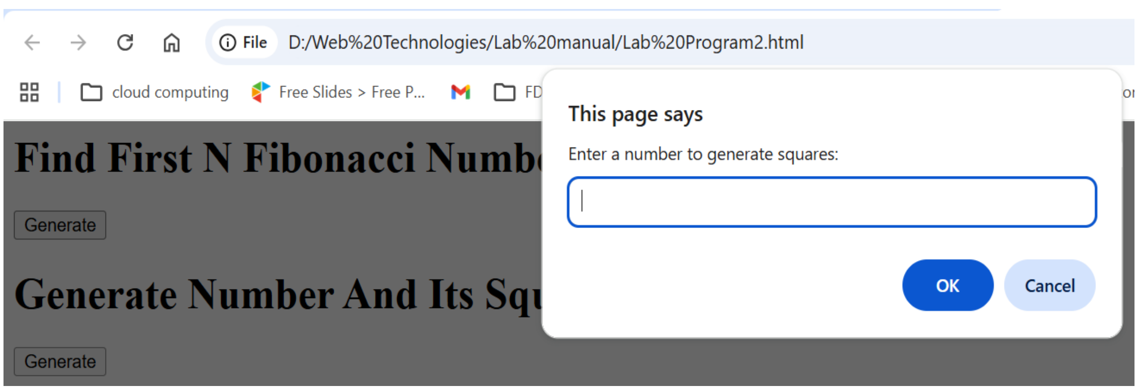
```
// Function to display squares table
    function squares( ) {
        var n = prompt("Enter a number for square table:", "5");
        n = parseInt(n);

        if (n <= 0) {
            alert("Please enter a positive number");
            return;
        }
        var output = "Number  Square\n";

        for (var i = 1; i <= n; i++) {
            output += i + "        " + (i * i) + "\n";
        }

        alert(output);
    }
  </script>
</head>

<body>
  <h1>JavaScript Demo</h1>

  <form action="#">
    <input type="button" value="Fibonacci Series" onclick="fibonacci()" />
    <br /><br />
    <input type="button" value="Squares Table" onclick="squares()" />
  </form>

</body>
</html>
```

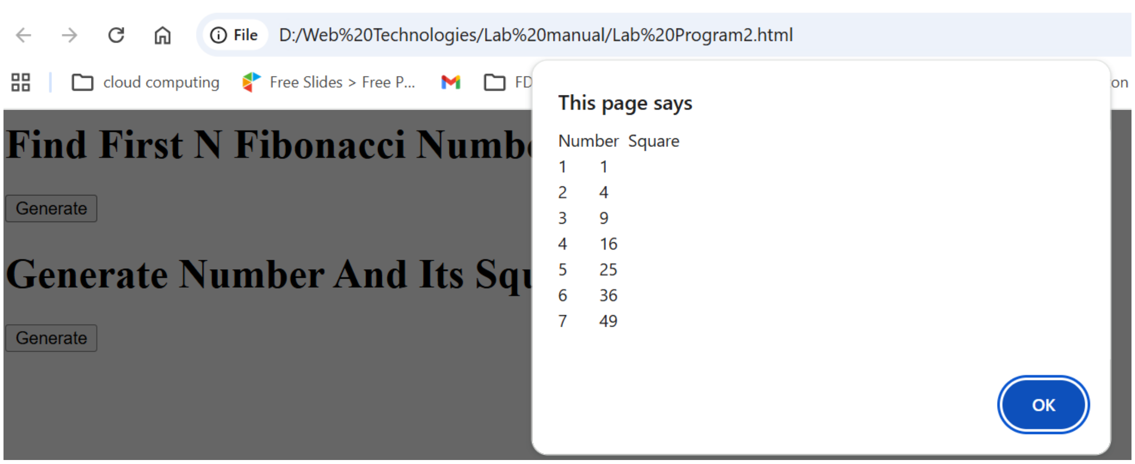
Output:




File
D:/Web%20Technologies/Lab%20manual/Lab%20Program2.html
cloud computing
Free Slides > Free P...
Find First N Fibonacci Numb
Generate
Generate Number And Its Squ
Generate
This page says
Enter the number of Fibonacci numbers to generate:
OK
Cancel


File
D:/Web%20Technologies/Lab%20manual/Lab%20Program2.html
cloud computing
Free Slides > Free P...
Find First N Fibonacci Numb
Generate
Generate Number And Its Squares
Generate
This page says
Fibonacci Series: 0 1 1 2 3
OK


File
D:/Web%20Technologies/Lab%20manual/Lab%20Program2.html
cloud computing
Free Slides > Free P...
Find First N Fibonacci Numb
Generate
Generate Number And Its Squ
Generate
This page says
Enter a number to generate squares:
OK
Cancel


File
D:/Web%20Technologies/Lab%20manual/Lab%20Program2.html
cloud computing
Free Slides > Free P...
Find First N Fibonacci Numb
Generate
Generate Number And Its Squ
Generate
This page says
Number Square
1 1
2 4
3 9
4 16
5 25
6 36
7 49
OK

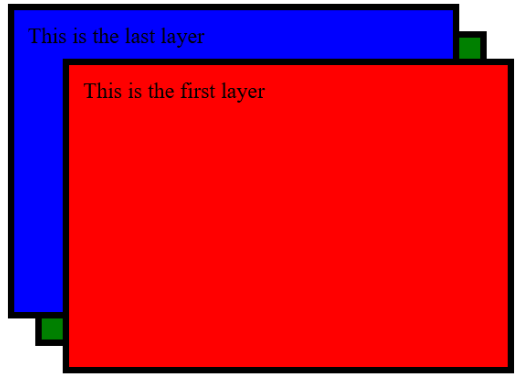
Program 3

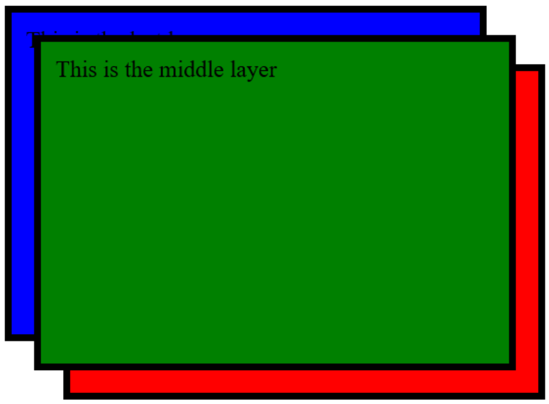
Develop and demonstrate, using JavaScript script, a XHTML document that contains three short paragraphs of text, stacked on top of each other, with only enough of each showing so that the mouse cursor can be placed over some part of them. When the cursor is placed over the exposed part of any paragraph, it should rise to the top to become completely visible. Modify the above document so that when a text is moved from the top stacking position, it returns to its original position rather than to the bottom.

```
<!DOCTYPE html>
<html>
<head>
<title>Stack Ordering</title>
<style type="text/css">
      .layer {
      border: solid thick black;
      padding: 10px;
      width: 300px;
      height: 200px;
      position: absolute;
      z-index: 0;
      }
      #layer1 { background-color: red; top: 140px; left: 240px; }
      #layer2 { background-color: green; top: 120px; left: 220px; }
      #layer3 { background-color: blue; top: 100px; left: 200px; }
</style>
</head>
<body>
    <p id="layer1" class="layer" onmouseover="mover(layer1);">This is the first layer</p>
    <p id="layer2" class="layer" onmouseover="mover(layer2);">This is the middle layer</p>
    <p id="layer3" class="layer" onmouseover="mover(layer3);">This is the last layer</p>
    <script type="text/javascript">

      var topLayer = document.getElementById('layer3');
      function mover(toTop)
      {
          topLayer.style.zIndex = "0";                  // Lower the current top layer
          toTop.style.zIndex = "1";                     // Bring the new one to top
          topLayer = toTop;                             // Update topLayer reference
      }
      </script>
      </body>
      </html>
```

Output:


This is the last layer


This is the last layer
This is the first layer

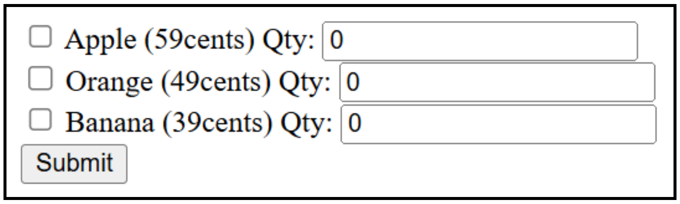

This is the middle layer

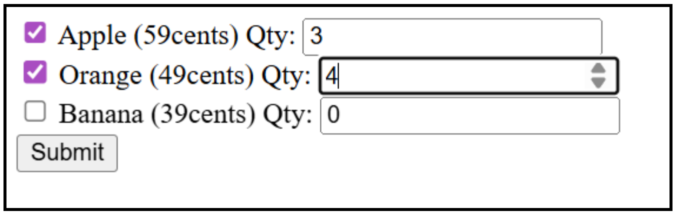
Program 4

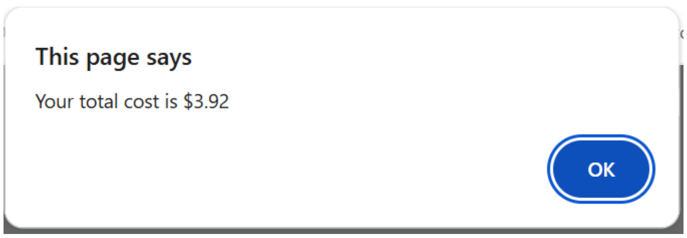
Develop, test and validate an XHTML document that has checkboxes for apple (59 cents each), orange (49 cents each), and banana (39 cents each) along with submit button. Each check box should have its own onclick event handler. These handlers must add the cost of their fruit to the total cost. An event handler for the submit button must produce an alert window with the message „your total cost is $xxx‟, where xxx is the total cost of the chose fruit, including 5 percent sales tax. This handler must return „false‟ (to avoid actual submission of the form data). Modify the document to accept quantity for each item using textboxes.

```
<!DOCTYPE html>
<html lang="en">

<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Fruit Cost Calculator</title>
</head>

<body>
   <label><input type="checkbox" id="apple"> Apple (59cents) Qty:
  <input type="number" id="apple_qty" min="0" value="0"></label><br>
   <label><input type="checkbox" id="orange"> Orange (49cents) Qty:
   <input type="number" id="orange_qty" min="0" value="0"></label><br>
   <label><input type="checkbox" id="banana"> Banana (39cents) Qty:
   <input type="number" id="banana_qty" min="0" value="0"></label><br>
   <button onclick="calculateTotal()">Submit</button>

   <script>
       function calculateTotal( )
       {
           let prices = { apple: 0.59, orange: 0.49, banana: 0.39 }, total = 0;
           for (let fruit in prices)
           {
               if (document.getElementById(fruit).checked)
              {
                 total += prices[fruit] * (parseInt(document.getElementById(fruit + "_qty").value) || 0);
              }
           }
           alert(`Your total cost is $${(total * 1.05).toFixed(2)}`);
       }

//When you want to add 5% to a total amount, you're increasing the original value by 5%. In math, this is
 //done by multiplying the total by (100% + 5%) = 105%.
//But JavaScript (and programming in general) doesn't use percentages directly. Instead, we use decimals:
//100% = 1.00
//5% = 0.05
//So, 105% = 1.05
 </script>
</body>
</html>
```

Output:


Apple (59cents) Qty: 0
Orange (49cents) Qty: 0
Banana (39cents) Qty: 0
Submit


Apple (59cents) Qty: 3
Orange (49cents) Qty: 4
Banana (39cents) Qty: 0
Submit


This page says
Your total cost is $3.92
OK